Sheet (2)

(1) Write a program that reads a Celsius degree in a double value from the console, then converts it to Fahrenheit and displays the result. The formula for the conversion is as follows:

$$\text{fahrenheit} = \left(\frac{9}{5}\right) \times \text{celsius} + 32$$

(2) Write a program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:

$$Area = (\text{Radius})^2 \times \pi$$

$$Volume = Area \times Length$$

(3) Write a program that prompts the user to enter a number of days. Calculate and display the number of years, months and days for the input. For simplicity, assume a year has 365 days.

(4) Write a program that prompts the user to enter two points $(x_1 + y_1)$ and $(x_2 + y_2)$ and displays their distance between them. The formula for computing the distance is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(5) Write a program that prompts the user to enter the coordinates of two points $(x_1 + y_1)$ and $(x_2 + y_2)$ and displays the slope of the line that connects the two points. The formula of the slope is $(y_2 - y_1)/(x_2 - x_1)$.

(6) A regular polygon is an n-sided polygon in which all sides are of the same length and all angles have the same degree (i.e., the polygon is both equilateral and equiangular). The formula for computing the area of a regular polygon is expressed as $A = \dfrac{n \times s^2}{4 \times \tan(\pi/n)}$ where s is the length of a side.

Write a C++ program that prompts the user to enter the number of sides and their length of a regular polygon and displays its area.

(7) Write a C++ program that reads three edges of a triangle and computes the area if the input is valid. Otherwise, it displays that the input is invalid. The formula for computing the area of a triangle is

$$s = (side1 + side2 + side3)/2$$

$$area = \sqrt{s(s - side1)(s - side2)(s - side3)}$$

(8) Write a program that prompts the user to enter the distance to drive, the fuel efficiency of the car in 100 km per liter, and the price in pounds per liter, and displays the fuel cost of the trip per pounds. Note that the range of fuel efficiency for small vehicles is 6:12.

(9) The two roots of a quadratic equation $ax^2 + bx + c = 0$ can be obtained using the following formula: $r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $b^2 - 4ac$ is called the discriminant of the quadratic equation. If it is positive, the equation has two real roots. If it is zero, the equation has one root. If it is negative, the equation has no real roots.

Write a C++ program that prompts the user to enter values for a, b, and c and displays the result based on the discriminant. If the discriminant is positive, display two roots. If the discriminant is 0, display one root. Otherwise, display "The equation has no real roots."

(10) Write a C++ program that prompts the user to enter three integers and display these integers in descending order.

(11) Write a C++ program that prompts the user to enter the three edges of a triangle and classify it as an equilateral, isosceles or a scalene triangle. First, check the data for a valid triangle, then compare the edges. If all edges are equal, display Equilateral Triangle; if only two edges are equal, display Isosceles Triangle and if none of the edges are equal, display Scalene Triangle.

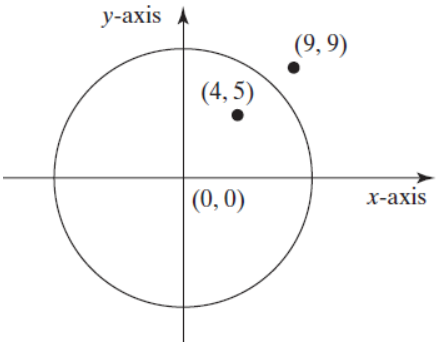
(12) Write a program that prompts the user to enter three points $(x1, y1), (x2, y2), (x3, y3)$ of a triangle and displays its area. The formula for computing the area of a triangle is

$$s = (side1 + side2 + side3)/2$$

$$area = \sqrt{s(s - side1)(s - side2)(s - side3)}$$

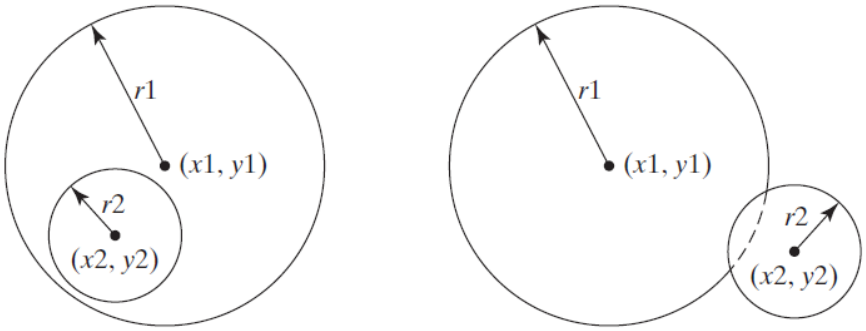
(13) Write a C++ program that prompts the user to enter a three-digit integer. The program displays the sum of all digits in the integer if the input is valid; otherwise, it displays a message indicating that the integer is not a three-digit number and hence, is invalid.

(14) Write a program that prompts the user to enter a point (x, y) and checks whether the point is within the circle centered at $(0, 0)$ with radius 10. For example, $(4, 5)$ is inside the circle and $(9, 9)$ is outside the circle, as shown in in the below figure.



(**Hint:** A point is in the circle if its distance to $(0, 0)$ is less than or equal to 10. The formula for computing the distance is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Test your program to cover all cases.)

(15) Write a program that prompts the user to enter the center coordinates and radii of two circles and determines whether the second circle is inside the first or overlaps with the first, as shown in the below figure.



(**Hint:** circle2 is inside circle1 if the distance between the two centers $\leq |r1 - r2|$ and circle2 overlaps circle1 if the distance between the two centers $\leq |r1 + r2|$. Test your program to cover all cases.)

(16) Write a program to display the current time in your location in the format hour:minute:second, using a 12-hour clock. Enter the time zone offset to Greenwich Mean Time (GMT) to adjust the time in the specified time zone.

Hint: Use the time(0) function, in the ctime header file, returns the current time in seconds elapsed since the time 00:00:00 on January 1, 1970 GMT.

(17) Write a C++ program that prompts the user to enter the coordinates of two points (x_1, y_1) and (x_2, y_2), and displays the line equation in the slope-intercept form, i.e., $y = mx + b$. m and b can be computed using the following formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \qquad \text{and} \qquad b = y_1 - mx_1$$

Don't display m if it is 1 and don't display b if it is 0.

(18) Body Mass Index (BMI) is a screening tool that estimates body fat based on a person's weight and height. It is calculated by dividing weight (in kilograms) by height squared (in meters). That is

$$BMI = \frac{Weight\ in\ Kg}{(Height\ in\ m)^2}$$

Write a C++ program that prompts the user to enter a person's weight in kilograms and height in centimeters. The program calculates the BMI and gives an interpretation for the given data according the following table;

BMI	Interpretation
$BMI < 18.5$	Underweight
$18.5.0 \leq BMI < 25.0$	Normal
$25.0 \leq BMI < 30.0$	Overweight
$30.0 \leq BMI$	Obese